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**HOSPITAL AND
CLINICAL PHARMACY**



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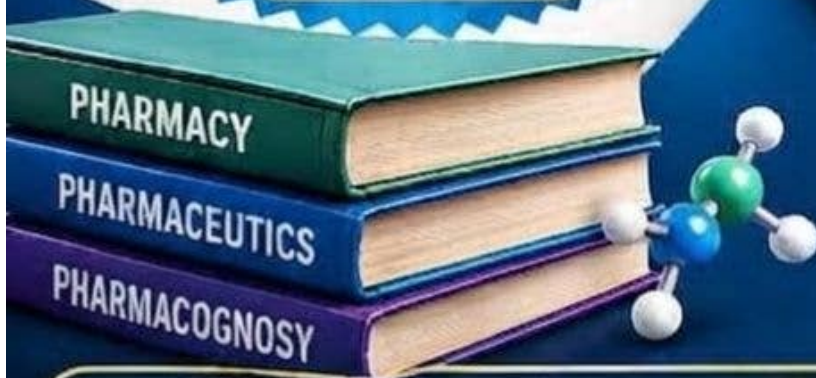


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As per the
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**HOSPITAL & CLINICAL
PHARMACY**

This note is
applicable only for
the 2027 (2nd year)
Academic Session

2nd year Suggestive Note
Session:- 2025-2027

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CHAPTER-1**HOSPITAL PHARMACY****Summary**

This chapter defines a hospital as a complex organization providing healthcare through specialized equipment and trained staff. According to the World Health Organization (WHO), hospitals serve as reservoirs of critical resources and knowledge, classified by the interventions they provide and their specific roles in the health system. Their core functions include improving medical standards, detecting diseases early, evaluating facility needs and serving as a bridge between the public and policymakers. Modern hospitals are classified based on the length of patient stay, clinical basis, ownership, size and system of medicine. Hospital pharmacy is a specialized healthcare service focused on selecting, preparing, storing and dispensing medicines and medical devices. Its primary objectives are to ensure the availability of high-quality, affordable

medications and to provide expert drug information to both healthcare professionals and patients. The scope of this department extends from managing drug supplies and inventory to manufacturing sterile preparations when they are not commercially available. While the pharmacist's role was traditionally limited to manufacturing and dispensing, it has evolved internationally toward patient-centered care, including education and clinical advice. A hospital pharmacist must possess a recognized degree (B.Pharm, M.Pharm or Pharm.D) and complete mandatory training and registration. Their responsibilities are divided between in-patient and out-patient areas, covering tasks such as maintaining medication histories, monitoring drug therapy effectiveness and participating in cardiopulmonary emergencies. To function effectively, they must maintain strong inter-professional relationships with doctors and nurses, ensuring rational drug therapy and safe administration. Furthermore, pharmacists must adhere to Good Pharmacy Practice (GPP) standards, which prioritize patient welfare and the promotion of cost-effective, rational prescribing.

Important Abbreviation

PCI	Pharmacy Council of India
M.Pharm	Masters of Pharmacy
Pharm.D	Doctor of Pharmacy
DCGI	Drugs Controller General of India
GPP	Good Pharmacy Practice
FIP	International Pharmaceutical Federation
NQAS	National Quality Assurance Standards
SQAC	State Quality Assurance Committee
SQAU	State Quality Assurance Unit
SISC	State Family Planning Indemnity Sub-committee
DQT	District Quality Team
DQAU	District Quality Assurance Unit
QCI	Quality Council of India
NDPS	Narcotic Drugs and Psychotropic Substances
CDSCO	Central Drugs Standard Control Organisation

Multiple Choice Questions

1. Q. According to WHO, hospitals are described as:

- a) Retail outlets for medicines
- b) Reservoirs of critical resources and knowledge
- c) Administrative centers for doctors
- d) Manufacturing units for drugs

Indrani Pathsala Answer: b) Reservoirs of critical resources and knowledge

2. Q. A "Long-term" hospital is classified as one where the patient stay is:

- a) Less than 30 days
- b) Exactly 30 days
- c) More than 30 days
- d) More than 1 year

Indrani Pathsala Answer: c) More than 30 days

3. Q. Which degree is NOT mentioned as a requirement to become a Hospital Pharmacist?

- a) B.Pharm
- b) M.Pharm
- c) Pharm.D
- d) D.Sc (Doctor of Science)

Indrani Pathsala Answer: d) D.Sc

4. Q. The minimum number of pharmacists required for a hospital with up to 200 beds is:

- a) 3
- b) 5
- c) 8
- d) 10

Indrani Pathsala Answer: c) 8

5. Q. In the organizational structure, the Head of Pharmacy Services (M.Pharm) reports directly to the:

- a) Nursing Superintendent b) Administrator c) Staff Pharmacist d) Local Body

Indrani Pathsala Answer: b) Administrator

6. Q. The "Cold Storage Area" in a hospital pharmacy is primarily used for:

- a) Storing surgical items b) Storing narcotic drugs
c) Storing thermolabile medicines d) Storing office records

Indrani Pathsala Answer: c) Storing thermolabile medicines

7. Q. Which committee should a pharmacist discuss issues with before advising medical staff?

- a) Sports Committee b) Pharmacy and Therapeutics Committee
c) Budget Committee d) Disciplinary Committee

Indrani Pathsala Answer: b) Pharmacy and Therapeutics Committee

8. Q. "Good Pharmacy Practice" (GPP) standards were first created by:

- a) PCI b) AICTE c) FIP (International Pharmaceutical Federation) d) DCGI

Indrani Pathsala Answer: c) FIP

9. Q. The primary objective of Hospital Pharmacy is to ensure medications are available at:

- a) High cost b) Affordable cost c) No cost only d) Market rate only

Indrani Pathsala Answer: b) Affordable cost

10. Q. Narcotic and alcohol records must be maintained by the hospital pharmacy under which scope?

- a) Manufacture b) Record Maintenance c) Research d) Education

Indrani Pathsala Answer: b) Record Maintenance

Short Answer Questions

1. Q. Define Hospital Pharmacy.

Indrani Pathsala Answer: It is a healthcare service involving the choosing, preparing, storing, compounding and dispensing of medicines and medical devices, while advising healthcare professionals and patients on their safe use.

2. Q. Mention any two functions of a hospital.

Indrani Pathsala Answer: 1) Reducing the occurrence of diseases through early detection. 2) Developing an effective system of clinical and administrative records.

3. Q. Where should a hospital pharmacy ideally be located?

Indrani Pathsala Answer: It should be situated on the ground floor or first floor to ensure easy accessibility for various departments and nursing stations.

4. Q. What is the minimum training period required post-degree for registration as per PCI?

Indrani Pathsala Answer: The candidate should undergo a training period of at least one year.

5. Q. State one administrative responsibility of a hospital pharmacist.

Indrani Pathsala Answer: Planning and implementing pharmacy policies, budgeting and maintaining stock records.

Fill in the Blanks

1. Q. A hospital with _____ beds is classified as a Medium Hospital.

Indrani Pathsala Answer: 501-1000

2. Q. The _____ is the head of the pharmacy department.

Indrani Pathsala Answer: Chief Hospital Pharmacist

3. Q. Under GPP, the pharmacist's top priority should be _____.

Indrani Pathsala Answer: patient welfare

4. Q. For a hospital with up to 50 beds, the required number of pharmacists is _____.

Indrani Pathsala Answer: 3

5. Q. The pharmacist assists the physician in selecting the _____ and dose schedule.

Indrani Pathsala Answer: drug regimen

Long Questions

1. Q. What are the functions of Hospital? Write about the classification of Hospital. Write the differences between Ambulatory and Institution patient care services. Draw a chart organisation of Hospital.

Indrani Pathsala Answer:

➤ Functions of Hospital:

Patient Care and Treatment

The most fundamental function is providing **diagnostic, curative and rehabilitative** services. This includes:

- **Outpatient Services (OPD):** Treating patients who do not require an overnight stay.
- **Inpatient Services (IPD):** Managing acute or chronic illnesses through surgery, medication and nursing care.
- **Emergency Care:** Providing immediate medical attention for life-threatening conditions or accidents.

Public Health and Disease Prevention

Hospitals act as a hub for community health through:

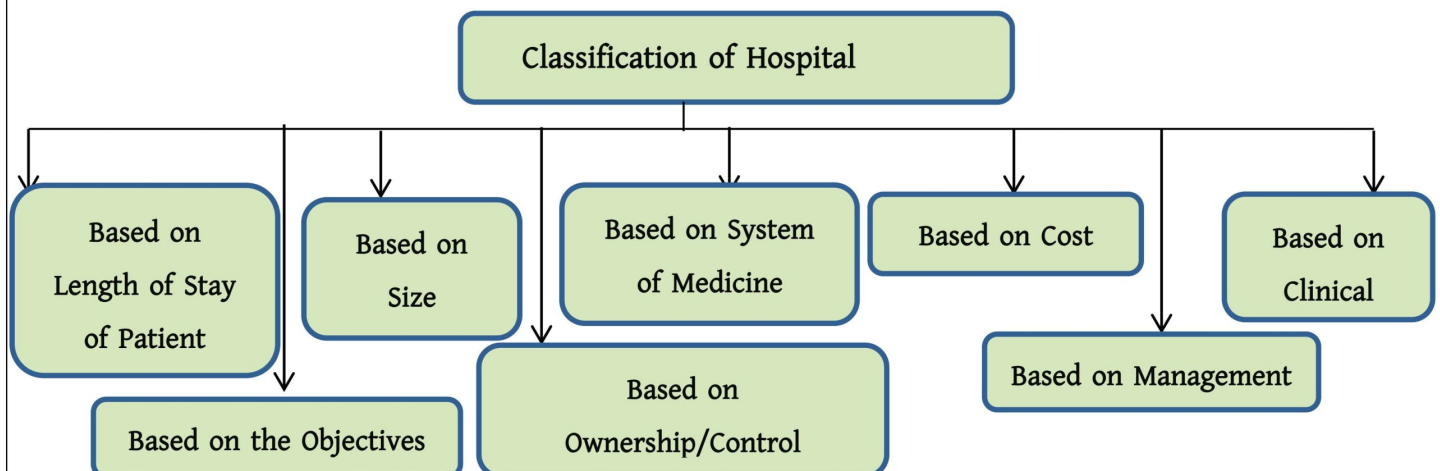
- **Preventive Services:** Organizing vaccination drives, health screenings and nutritional counseling.
- **Health Education:** Raising awareness about infectious diseases, family planning and lifestyle disorders (like diabetes or hypertension).

Training and Research

Hospitals serve as centers for professional development and scientific advancement:

- **Education:** Providing a practical training ground for medical students, nurses and pharmacists.
- **Medical Research:** Conducting clinical trials and studies to develop new treatments, improve surgical techniques and enhance drug safety.

➤ Classification of Hospital:

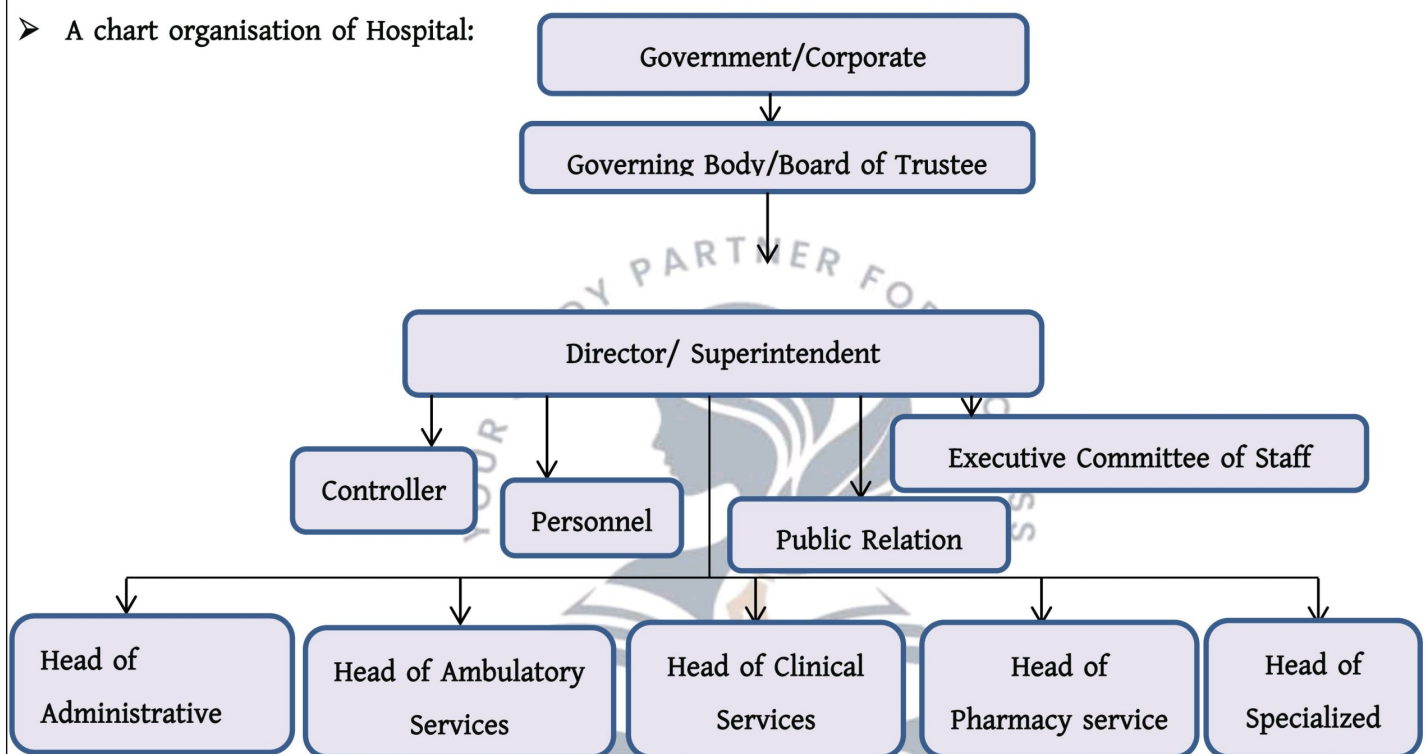


➤ Comparison Between Ambulatory and Institutional Patient Care

Feature	Ambulatory Care Services	Institutional (Inpatient) Care Services
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Definition	Medical services provided on an outpatient basis where patients return home the same day.	Healthcare provided to patients who are formally admitted to a facility for at least one night.
Duration	Short-term; usually lasts from a few minutes to several hours.	Long-term; involves an overnight stay or extended period of days/weeks.
Cost	Generally more affordable as it excludes room charges and 24/7 nursing fees.	More expensive due to costs associated with boarding, continuous monitoring and utilities.
Intensity of Care	Suitable for minor illnesses, routine check-ups and diagnostic tests.	Required for complex surgeries, severe trauma or acute life-threatening conditions.
Examples	Doctor's clinics, pharmacies, urgent care centers and diagnostic labs.	General hospitals, specialized surgical centers and nursing homes.

➤ A chart organisation of Hospital:



2. What are the differences between Hospital Pharmacy and Community Pharmacy. Write any four points of Supporting Services and 2 points of the level of caring health care in Hospital.

Answer:

➤ Hospital Pharmacy vs. Community Pharmacy

Feature	Hospital Pharmacy	Community Pharmacy
Setting	Operates within a hospital, clinic or nursing home.	Operates as a retail store or a standalone local chemist shop.
Primary Focus	Focuses on inpatient care , specialized medication and clinical rounds.	Focuses on outpatient care , retail sales and public health.
Patient Interaction	Indirect; pharmacists usually interact with doctors and nurses.	Direct; pharmacists interact directly with the general public/customers.
Stock & Scope	Stocks specialized drugs (IV fluids, oncology drugs) and handles complex dosing.	Stocks common medications, OTC (over-the-counter) products and daily essentials.

➤ Supporting Services and the level of caring health care in Hospital:

Supporting Services :

Supporting services (also known as ancillary services) provide the necessary infrastructure and clinical assistance required for patient care:

- **Pharmacy Services:** Responsible for the procurement, storage and dispensing of medications to both inpatients and outpatients.
- **Diagnostic Services:** Includes **Radiology** (X-rays, CT scans) and **Pathology** (blood and urine tests) to help doctors identify diseases.
- **Dietary Services:** Ensures patients receive nutritious meals tailored to their specific medical conditions (e.g., diabetic or low-sodium diets).
- **Central Sterile Supply Department (CSSD):** Responsible for cleaning, disinfecting and sterilizing all medical instruments and equipment to prevent hospital-acquired infections.

Levels of Caring Health Care:

Health care levels are categorized based on the complexity of the medical cases and the resources required:

- **Secondary Care:** This is the first level of referral. It involves medical specialists (like cardiologists or urologists) who handle more specific health problems that cannot be managed at the primary health center level.
- **Tertiary Care:** This is specialized consultative care, usually for inpatients and on referral from primary or secondary health professionals. It involves advanced diagnostic support and complex surgeries (e.g., neurosurgery or transplant units) in a large hospital setting.

3. **Q. What is Hospital Pharmacy? Write the scope, national and international scenario, function, objective and developments in Pharmacy education of Hospital Pharmacy.**

Indrani Pathsala Answer:

➤ **Hospital Pharmacy:**

Hospital Pharmacy is a specialized department within a healthcare institution (such as a hospital or clinic) responsible for the procurement, storage, testing, compounding and dispensing of medications to patients under medical supervision. Its primary goal is to ensure the safe and effective use of drugs for both inpatients and outpatients.

➤ **Scope of Hospital Pharmacy**

The scope is rapidly expanding from traditional drug dispensing to clinical involvement:

- **Clinical Services:** Participation in ward rounds, patient medication history reviews and therapeutic drug monitoring.
- **Specialized Dispensing:** Handling complex medication needs, such as **Total Parenteral Nutrition (TPN)** and **Intravenous (IV) admixtures**.
- **Drug Information:** Providing unbiased information about drugs to physicians, nurses and patients.
- **Manufacturing:** Small-scale production of sterile and non-sterile preparations that are not commercially available.
- **Pharmacovigilance:** Monitoring and reporting **Adverse Drug Reactions (ADRs)** within the hospital.

National & International Scenario

- **National (India):** The scenario has evolved with the implementation of **PCI (Pharmacy Council of India) 2020** regulations. There is a shift toward "Clinical Pharmacy," where Pharm.D and M.Pharm graduates are increasingly integrated into the healthcare team. However, infrastructure in rural government hospitals still faces challenges.

- **International:** In developed nations (USA, UK), hospital pharmacists have **prescribing rights** and are primary members of the multidisciplinary team. Technology, such as robotic dispensing and computerized physician order entry (CPOE), is standard.

Functions of Hospital Pharmacy

- **Procurement and Inventory:** Selecting reliable suppliers and maintaining an uninterrupted supply of medicines.
- **Storage:** Ensuring drugs are stored under proper conditions (e.g., cold chain for vaccines).
- **Dispensing:** Providing medications for both inpatients (via ward boxes or unit-dose systems) and outpatients.
- **Drug Education:** Training nurses and medical staff on the correct administration of new drugs.

Objectives

- To ensure the **availability** of the right drug, for the right patient, in the right dose, at the right time.
- To minimize **medication errors** through double-check systems.
- To ensure **cost-effective** treatment by following the hospital formulary.
- To promote the **Rational Use of Drugs**.

Developments in Pharmacy Education

The education system has transitioned from a product-oriented approach to a **patient-oriented approach**:

- **Revised Curriculum:** Implementation of the **ER-2020** for D.Pharm and specialized B.Pharm/M.Pharm courses focuses more on hospital internships and clinical exposure.
- **Introduction of Pharm.D:** This 6-year doctoral program was introduced to bridge the gap between pharmacists and physicians.
- **Continuing Education:** Mandatory credits and workshops for registered pharmacists to keep them updated on new drug discoveries and healthcare technologies.
- **Digital Learning:** Use of simulation software and digital health records in training to prepare students for modern smart hospitals.

4. Q. Discuss the location, layout, facilities required, equipment required for Hospital Pharmacy.

Indrani Pathsala Answer:

Location of Hospital Pharmacy

The location is strategic and depends on the type of patients served:

- **Outpatient Pharmacy:** Should be located on the **ground floor** near the entrance or the main registration area for easy accessibility.
- **Inpatient Pharmacy:** Can be centrally located or decentralized (floor-wise) to ensure quick delivery of medicines to the wards.
- **Store/Warehouse:** Usually located in the basement or a less busy area but must have a direct lift or ramp access for receiving bulk supplies.

Layout of Hospital Pharmacy

The layout should follow a logical flow (Unidirectional flow) to prevent cross-contamination and errors. Key areas include:

- **Reception/Waiting Area:** Sufficient space for patients/relatives with a token system.
- **Dispensing Area:** Provided with counters, racks and drawers for fast-moving drugs.
- **Compounding/Manufacturing Area:** A separate, clean room for preparing non-sterile or sterile products.
- **Storage Area:** Divided into general storage, **Cold Room** (for vaccines/insulin) and a **Narcotic Vault** (locked area for controlled substances).

- **Administrative Area:** Office space for the Head of Pharmacy and record-keeping.

Facilities Required

To function effectively, the pharmacy must have:

- **Climate Control:** Air conditioning and ventilation to maintain drug stability.
- **Lighting:** Well-lit areas to prevent dispensing errors.
- **Water Supply:** Continuous supply of potable and distilled water.
- **Sanitation:** Proper drainage and waste disposal systems (as per Biomedical Waste Management rules).
- **Communication:** Intercoms, computers and internet for prescription processing and inventory management.

Equipment Required

Equipment is categorized based on the function of the pharmacy:

Category	Essential Equipment
Storage	Refrigerators (2°C to 8°C), Deep Freezers, Cooling Cabinets and Racks/Pallets.
Dispensing	Counting trays, spatulas, labels and prescription containers.
Compounding	Mortar and pestles, balances (Electronic/Analytical), Ovens and Hot plates.
Sterile Area	Laminar Air Flow (LAF), Autoclaves and Membrane filters.
Quality Control	pH meters, Viscometers and basic testing kits.
Safety	Fire extinguishers, Eye-wash stations and First-aid kits.

Ideal Layout Diagram (Conceptual):

[Entrance]→[Waiting Area]→[Dispensing Counter]→[Compounding/Work Bench]→[Storage/Cold Chain]→[Admin Office]

Conclusion:

A well-planned location and layout minimize the distance travelled by staff, reduce the "turnaround time" for prescriptions and ensure that medicines are stored under conditions that maintain their potency and safety.

5. Q. Who is a Hospital Pharmacist? What are the Professional responsibilities and job specifications of Hospital Pharmacist?

Indrani Pathsala Answer:

➤ Hospital Pharmacist:

A **Hospital Pharmacist** is a legally qualified and registered professional who is responsible for the management, preparation and dispensing of medications within a hospital or clinical setting, ensuring their safe and effective use for patients.

➤ Professional Responsibilities of a Hospital Pharmacist

The responsibilities can be categorized into administrative, clinical and advisory roles:

- **Dispensing and Distribution:** Ensuring the accurate supply of medications to various wards (Inpatients) and to the outpatient department (OPD). This includes managing unit-dose systems and floor stock.
- **Inventory Management:** Responsible for the procurement, scientific storage (maintaining cold chains) and inventory control (using techniques like ABC or VED analysis) of drugs and surgical supplies.
- **Clinical Pharmacy Services:** Participating in medical rounds, taking patient medication histories and monitoring drug therapy for potential interactions or contraindications.
- **Compounding and Admixtures:** Preparing specialized medications not available commercially, such as IV admixtures, total parenteral nutrition (TPN) and cytotoxic (cancer) drugs in a sterile environment.

- **Drug Information Services:** Acting as a consultant to doctors and nurses regarding drug dosages, side effects and new drug availability.
- **Patient Counseling:** Educating patients on how to use their medications, especially specialized devices like inhalers or insulin pens, to ensure compliance.
- **Pharmacovigilance:** Identifying, documenting and reporting **Adverse Drug Reactions (ADRs)** to the appropriate authorities.
- **Pharmacy and Therapeutics Committee (PTC):** Serving as a key member or secretary of the PTC to help develop the Hospital Formulary and drug use policies.

Job Specifications of a Hospital Pharmacist

Job specifications refer to the minimum requirements a candidate must meet to hold the position:

- **Educational Qualification:**
 - Minimum: **Diploma in Pharmacy (D.Pharm)** for entry-level roles.
 - Preferred for Clinical Roles: **Bachelor of Pharmacy (B.Pharm), Master of Pharmacy (M.Pharm)** in Pharmacy Practice or **Pharm.D.**
- **Legal Requirement:** Must be a **Registered Pharmacist** under the State Pharmacy Council (Pharmacy Act 1948).
- **Technical Skills:**
 - Knowledge of pharmacology, toxicology and drug interactions.
 - Ability to read and interpret complex prescriptions.
 - Proficiency in pharmacy management software and digital record-keeping.
- **Soft Skills:**
 - Excellent communication skills for patient counseling and coordinating with medical staff.
 - High level of accuracy and attention to detail to prevent medication errors.
 - Ethical integrity and ability to work under pressure in emergency situations.

6. Q. Shortly explain about GPP and NQAS Guidelines and Hospital Pharmacy Standards.

Indrani Pathsala Answer:

➤ Good Pharmacy Practice (GPP)

GPP is the practice of pharmacy that responds to the needs of the people who use the pharmacists' services to provide optimal, evidence-based care.

Core Pillars of GPP:

- **Health Promotion:** Pharmacists must participate in health screening and public health campaigns (e.g., tobacco cessation or immunization).
- **Supply Chain Management:** Ensuring the integrity of the medicine supply chain from procurement to the patient.
- **Patient Counseling:** Providing clear instructions on how to take medication, potential side effects and storage.
- **Professionalism:** Maintaining an updated knowledge base and adhering to ethical standards.

GPP Requirements in Hospital Pharmacy:

- **Selection:** Preparing a hospital formulary based on efficacy and safety.
- **Storage:** Ensuring drugs are kept in a secure, clean environment with temperature monitoring.
- **Dispensing:** Every prescription must be validated for clinical appropriateness before the medicine is handed over.

➤ National Quality Assurance Standards (NQAS)

NQAS was developed by the Ministry of Health and Family Welfare (India) to improve the quality of care in public health facilities (District Hospitals, CHCs, PHCs).

Key Areas of NQAS for Pharmacy:

- **Service Provision:** Availability of essential drugs 24/7.
- **Patient Rights:** Maintaining patient confidentiality and providing information in local languages.
- **Infection Control:** Proper disposal of pharmaceutical waste and maintaining hand hygiene in the dispensing area.
- **Quality Management:** Periodic internal audits and "Patient Satisfaction Surveys" to improve services.

The "Eight Areas" of NQAS Concern:

NQAS evaluates the pharmacy based on:

- **Inventory Management:** Use of software for stock tracking.
- **Prescription Audit:** Checking for polypharmacy and irrational drug use.
- **Look-Alike Sound-Alike (LASA):** Strict segregation of confusing drug names.
- **Expiry Management:** Using the FEFO (First Expiry, First Out) method.

➤ Hospital Pharmacy Standards

Hospital Pharmacy Standards are the baseline requirements for a pharmacy to function legally and safely within a medical institution. These are governed by the **Pharmacy Council of India (PCI)** and the **Drugs and Cosmetics Act**.

A. Legal and Administrative Standards

- **Licensing:** The hospital must possess valid drug licenses (Forms 20 and 21).
- **Qualified Staff:** The pharmacy must be headed by a Registered Pharmacist.
- **Space:** Minimum space requirements must be met (e.g., 10 square meters for a retail/hospital pharmacy).

B. Operational Standards

- **Standard Operating Procedures (SOPs):** Written protocols for every task, from cleaning floors to handling narcotics.
- **Documentation:** Maintaining registers for scheduled drugs (Schedule X/H), purchase records and patient profiles.
- **Emergency Medications:** Ensuring "Crash Carts" in the ICU/ER are always stocked with life-saving drugs.

C. Clinical Standards

- **Therapeutic Drug Monitoring (TDM):** Monitoring drugs with a narrow therapeutic index (e.g., Digoxin, Phenytoin).
- **Medication Error Reporting:** A non-punitive system for staff to report near-misses or errors to improve the system.

Comparison Summary

Feature	GPP	NQAS	Hospital Standards
Origin	FIP / WHO	Govt. of India (MoHFW)	PCI / Legal Acts
Focus	Professional behavior and patient care.	Institutional quality and public health efficiency.	Minimum legal and safety requirements.
Primary Goal	Optimal therapeutic outcomes.	Certification and standardized care.	Legal compliance and basic safety.

7. Q. Discuss about NABH, Accreditation and role of Pharmacist.

Indrani Pathsala Answer:

NABH (National Accreditation Board for Hospitals)

- **Definition:** NABH is a constituent board of the **Quality Council of India (QCI)**, established to set benchmarks for the quality and safety of healthcare providers in India.
- **Standards:** It operates based on a set of rigorous standards (currently the 5th Edition) that cover patient care, medication management and hospital infection control.
- **Global Recognition:** It is an institutional member of the **ISQua** (International Society for Quality in Health Care), meaning NABH-accredited hospitals meet international quality benchmarks.

Accreditation

- **Purpose:** Accreditation is a voluntary process where a healthcare organization is evaluated by an external body to ensure it meets pre-defined quality standards.
- **The Process:**
 - **Self-Assessment:** The hospital audits its own systems against NABH standards.
 - **Pre-Assessment:** An official gap analysis by NABH assessors.
 - **Final Assessment:** A detailed on-site evaluation of clinical and non-clinical services.
- **Outcome:** It ensures high-quality patient care, reduced risks (like infections or medical errors) and increases the hospital's credibility and "branding."

Role of the Pharmacist in NABH

The pharmacist is a central figure in the **Management of Medication (MOM)** chapter of the NABH standards.

Their key roles include:

- **Medication Safety:** Identifying and labeling **LASA (Look-Alike, Sound-Alike)** and **High-Risk medications** to prevent dispensing errors.
- **Formulary Management:** Ensuring that only medicines listed in the official Hospital Formulary are stocked and used.
- **Prescription Audit:** Regularly reviewing prescriptions for legibility, correct dosing and drug interactions to ensure rational drug use.
- **Storage & Inventory:** Maintaining proper storage conditions (like cold chain/refrigeration for vaccines) and using **FIFO (First-In, First-Out)** to prevent the use of expired drugs.
- **ADR Reporting:** Monitoring and documenting **Adverse Drug Reactions (ADRs)** and reporting them to the national pharmacovigilance program.
- **Staff Training:** Educating nursing and medical staff on safe drug administration, spill management (for hazardous drugs) and emergency protocols.

CHAPTER-2**DIFFERENT COMMITTEES IN THE HOSPITAL****Summary**

This chapter details the critical committees and systems within a hospital that ensure safe and rational drug therapy. The Pharmacy and Therapeutics Committee (PTC) is a multidisciplinary group of physicians and pharmacists that formulates policies on the therapeutic use of drugs and manages the hospital formulary. Its objectives are both advisory, such as suggesting drug distribution methods and educational such as managing staff training programs. The PTC plays a vital role in drug safety, adverse drug reaction (ADR) reporting and maintaining emergency drug lists. Within this committee, the pharmacist serves as a key administrator by creating schedules, recording minutes and ensuring the selection of affordable, effective medications.

The Hospital Formulary is a regularly revised list of medications stocked by the pharmacy, developed by the medical staff and PTC to ensure clinical utility and cost-effectiveness. It provides essential information on approved drug products, hospital policies and specific dosage schedules. While it helps lower inventory costs and enhances quality assurance, it may restrict a physician's choice of specific brands. The pharmacist is primarily responsible for the development, supply specifications and quality maintenance of the formulary.

Finally, the Infection Control Committee (ICC) focuses on preventing nosocomial (hospital-acquired) infections through surveillance, policy-making and antimicrobial stewardship. It is composed of various department heads, including the hospital director and chief nurse. Pharmacists support the ICC by managing anti-infective supplies, monitoring antibiotic usage statistics and educating staff on infection prevention. Crucially, pharmacists play a lead role in preventing antimicrobial resistance (AMR) by advocating for responsible antibiotic use and ensuring appropriate immunization practices.

Important Abbreviation

PTC	Pharmacy AND Therapeutics Committee
CAMs	Complementary and Alternative Medicines
ICC	Infection Control Committee
HF	Hospital Formulary

Multiple Choice Questions

1. Q. The Pharmacy and Therapeutics Committee (PTC) is primarily composed of:

- a) Only Pharmacists
- b) Physicians, Pharmacists and other health personnel
- c) Hospital Administrators and Patients
- d) Medical Representatives

Indrani Pathsala Answer: b) Physicians, Pharmacists and other health personnel

2. Q. Which of the following is an advisory objective of the PTC?

- a) To examine adverse drug reactions
- b) To offer hospital formulary system
- c) To help formulate professional policies regarding drug selection and usage
- d) To manage training programmes

Indrani Pathsala Answer: c) To help formulate professional policies regarding drug selection and usage

3. Q. The secretary of the Pharmacy and Therapeutics Committee is typically the:

- a) Hospital Director
- b) Chief Nurse
- c) Pharmacist
- d) Physician

Indrani Pathsala Answer: c) Pharmacist

4. Q. The "Hospital Formulary" is a list of:

- a) All drugs available in the global market
- b) Drugs that the hospital pharmacy stocks
- c) Only experimental drugs
- d) Hospital staff members

Indrani Pathsala Answer: b) Drugs that the hospital pharmacy stocks

5. Q. Which committee is responsible for the prevention and control of nosocomial infections?

- a) Pharmacy and Therapeutics Committee
- b) Budget Committee
- c) Infection Control Committee (ICC)
- d) Library Committee

Indrani Pathsala Answer: c) Infection Control Committee (ICC)

6. Q. A primary advantage of a Hospital Formulary is that it:

- a) Increases drug inventory costs
- b) Lowers drug inventory cost and improves procurement
- c) Allows physicians to prescribe any brand of their choice
- d) Secretly adds drugs of unknown composition

Indrani Pathsala Answer: b) Lowers drug inventory cost and improves procurement

7. Q. In an emergency drug box, which of the following is a standard drug requirement?

- a) Paracetamol syrup b) Atropine sulphate (0.4mg/ml) c) Vitamin C tablets d) Cough lozenges

Indrani Pathsala Answer: b) Atropine sulphate (0.4mg/ml)

8. Q. The chairperson of the Infection Control Committee (ICC) is usually the:

- a) Chief Nurse b) Hospital Director c) Head of Cleaning Department d) Staff Pharmacist

Indrani Pathsala Answer: b) Hospital Director

9. Q. Drug Utilisation Review (DUR) is an important function of the PTC to ensure:

- a) High profit for the pharmacy b) Safe and rational use of drugs
c) Using drugs past their expiry date d) Excluding all generic names

Indrani Pathsala Answer: b) Safe and rational use of drugs

10. Q. Antimicrobial resistance occurs when:

- a) Drugs become more expensive b) Microorganisms develop the ability to resist medications used to treat them
c) Patients refuse to take medications d) Vaccines are stored in the pharmacy

Indrani Pathsala Answer: b) Microorganisms develop the ability to resist medications used to treat them

Short Answer Questions

1. Q. Define Hospital Formulary.

Indrani Pathsala Answer: It is a method where the medical staff and PTC select and evaluate medicinal agents and dosage forms considered most useful in patient care, which the hospital pharmacy then stocks.

2. Q. State one educational objective of the PTC.

Indrani Pathsala Answer: To aid in policy making that satisfies the staff's demand for up-to-date information about drugs and their use.

3. Q. What is the role of the PTC regarding drug product defects?

Indrani Pathsala Answer: The PTC obtains information on defective drugs and informs the manufacturer/supplier or the FDA to take appropriate action.

4. Q. Mention two items that should be in an Emergency Box besides drugs.

Indrani Pathsala Answer: Syringes (various sizes) and needles.

5. Q. What is the pharmacist's role in preventing antimicrobial resistance?

Indrani Pathsala Answer: The pharmacist prevents antibiotic overuse by ensuring safe distribution and raising awareness of the risks of inappropriate antibiotic use.

Fill in the Blanks

1. Q. The PTC should ensure that _____ of drugs follows Good Manufacturing Practices (GMP).

Indrani Pathsala Answer: manufacturing

2. Q. Hospital Formulary should be revised _____ as new drugs are added or removed.

Indrani Pathsala Answer: annually

3. Q. The PTC transmits Adverse Drug Reaction (ADR) data to regulatory agencies such as the _____.

Indrani Pathsala Answer: FDA

4. Q. Poisons and poisonous materials must be stored _____ in the pharmacy and wards.

Indrani Pathsala Answer: separately

5. Q. In the ICC, the chief infection control team is typically found in the department of _____.

Indrani Pathsala Answer: microbiology

Long Questions

1. Q. What is PTC? Discuss about the role, objective, functions, composition, policy of PTC and Write the role of Pharmacist in PTC.

Indrani Pathsala Answer:

➤ PTC stands for the **Pharmacy and Therapeutics Committee**. It is a dynamic, multi-disciplinary advisory body within a hospital that serves as the connecting link between the medical staff and the pharmacy department.

➤ Here is a detailed discussion of its various aspects:

Objectives of PTC

The primary goals of the committee are:

- **Advisory:** To advise the medical staff and hospital administration on all matters related to the use of drugs.
- **Educational:** To formulate programs designed to meet the needs of professional staff (doctors, nurses and pharmacists) regarding drug knowledge.
- **Safety:** To ensure the highest quality of patient care by promoting the safe and rational use of medications.

Composition of PTC

The committee must include representatives from various healthcare sectors to be effective:

- **Chairman:** Usually a senior Physician/Medical Superintendent.
- **Secretary:** Usually the **Chief Pharmacist** (Head of Pharmacy).
- **Members:** * Representatives from major clinical departments (Surgery, Medicine, Gynecology, etc.).
 - A nursing representative (usually the Nursing Superintendent).
 - A representative from hospital administration.

Functions of PTC

The PTC acts as a watchdog for medication management:

- **Formulary Management:** Developing and maintaining the **Hospital Formulary** (the list of approved drugs for use in the hospital).
- **Drug Evaluation:** Evaluating new drugs based on their therapeutic merit, safety and cost before adding them to the formulary.
- **Policy Development:** Creating policies regarding drug distribution, emergency drug lists and the use of investigational drugs.
- **Adverse Drug Reaction (ADR) Monitoring:** Reviewing reports of ADRs occurring within the hospital.
- **Medication Error Review:** Analyzing errors in prescribing or dispensing to prevent future occurrences.

Policies of PTC

PTC policies are the "laws" of medication use in the hospital:

- **Automatic Stop Orders:** Policies that automatically cancel orders for high-risk drugs (like narcotics or antibiotics) after a specific period unless renewed by a doctor.
- **Generic Prescribing:** Promoting the use of generic names rather than brand names to reduce costs.
- **Drug Recall:** Establishing protocols for quickly removing defective or dangerous drugs from hospital shelves.
- **Emergency Drugs:** Defining which life-saving drugs must be available in "Crash Carts" at all times.

Role of the Pharmacist in PTC

The pharmacist (usually the Secretary) is the most active member of the committee. Their roles include:

- **Preparation of Agenda:** Organizing the topics to be discussed in the meetings.
- **Data Provider:** Presenting scientific data on drug efficacy, side effects and cost comparisons to help the committee make informed decisions.
- **Formulary Maintenance:** Updating the hospital drug list and ensuring all departments adhere to it.
- **Reporting:** Documentation of ADRs and medication errors to be reviewed by the committee.
- **Implementation:** Ensuring that the policies decided by the PTC are actually followed in the daily operations of the hospital pharmacy.
- **Minutes of Meeting:** Recording the minutes of each meeting and circulating them to the hospital management and staff.

2. Q. Write the definition of Hospital Formulary. Discuss the objectives, procedure for development, advantages and disadvantages guiding principles for use in Hospital Formulary and write the role of pharmacist in it. What are the differences between Private Formulary and National Formulary.

Indrani Pathsala Answer:

➤ **Definition**

A **Hospital Formulary** is a continually revised compilation of pharmaceuticals (plus important ancillary information) that reflects the current clinical judgment of the medical staff. It is essentially a "master list" of medications approved for use within a specific healthcare institution.

➤ **Objectives**

- **Therapeutic:** To ensure that only the most effective and safest drugs are available for patient care.
- **Economic:** To reduce drug costs by selecting cost-effective therapies and minimizing the number of redundant brands.
- **Educational:** To provide staff with updated information on drug dosages, side effects and administration.
- **Standardization:** To simplify the procurement, storage and distribution processes for the pharmacy.

Procedure for Development:

The development of a formulary is a systematic process managed by the PTC (Pharmacy and Therapeutics Committee):

- **Selection of Drugs:** Drugs are selected based on their therapeutic merit, safety and price.
- **Generic Names:** Drugs are usually listed by their non-proprietary (generic) names.
- **Monograph Preparation:** Each entry includes the drug's name, strength, dosage form, indications and side effects.
- **Review and Approval:** The medical staff reviews the proposed list and the PTC grants final approval.
- **Distribution:** The finalized formulary is printed or uploaded digitally for use by doctors, nurses and pharmacists.

Advantages and Disadvantages:

Advantages	Disadvantages
Safety: Reduces errors by limiting the variety of drugs staff must be familiar with.	Restricted Choice: Doctors may feel their "clinical freedom" to prescribe any drug is limited.
Cost Control: Bulk purchasing of fewer items leads to significant savings.	Delay in New Drugs: There might be a lag between the release of a new drug and its inclusion in the formulary.
Rational Use: Encourages evidence-based prescribing rather than following marketing trends.	Administrative Burden: Requires constant updating and monitoring by the PTC.

Guiding Principles for Use:

- **Medical Staff Acceptance:** The formulary must be accepted and followed by all clinical departments.
- **Continuous Revision:** It must be updated regularly to include new therapeutic breakthroughs.
- **Generic Preference:** The use of generic names is prioritized over brand names.
- **Non-Formulary Drugs:** A clear protocol must exist for obtaining a non-listed drug in special emergency cases.

Role of Pharmacist in Hospital Formulary

The pharmacist is the "architect" of the formulary:

- **Information Resource:** Providing the PTC with comparative data on drug efficacy and costs.
- **Secretary of PTC:** Organizing meetings and documenting the decisions regarding drug additions or deletions.
- **Quality Control:** Ensuring that the drugs procured meet the standards listed in the formulary.
- **Monitoring:** Auditing prescriptions to ensure doctors are adhering to the formulary guidelines.

➤ **Differences Between Private (Hospital) and National Formulary**

Feature	Private (Hospital) Formulary	National Formulary (e.g., NFI)
Scope	Restricted to a specific hospital or hospital group.	Applicable to the entire country.
Selection	Based on the specific needs and budget of that hospital.	Based on national health priorities and general disease prevalence.
Authority	Managed by the individual hospital's PTC.	Managed by a central government body (e.g., IPC in India).
Accessibility	Available to the staff of that institution.	Available to all healthcare professionals nationwide.

3. Q. What is ICC? Briefly explain composition, function and objectives of ICC. Mention the role of Pharmacist in ICC and the role of Pharmacist in preventing anti-microbial resistance.

Indrani Pathsala Answer:

➤ ICC stands for the **Infection Control Committee**. It is a specialized multi-disciplinary body in a hospital responsible for preventing and controlling **Healthcare-Associated Infections (HAIs)** (also known as nosocomial infections).

➤ **Composition of ICC**

To be effective, the ICC requires a team of experts from various departments:

- **Chairman:** Usually the Medical Superintendent or a Senior Physician.
- **Infection Control Officer:** Typically a **Microbiologist**.
- **Member Secretary:** The **Chief Pharmacist** or a Senior Clinician.
- **Infection Control Nurses (ICN):** Dedicated nurses who monitor wards for infection signs.
- **Members:** Representatives from Surgery, Nursing, Housekeeping and Hospital Administration.

Objectives of ICC

- **Patient Safety:** To minimize the risk of patients contracting new infections during their hospital stay.
- **Staff Protection:** To protect healthcare workers from occupational exposure to infections (e.g., needle-stick injuries).
- **Surveillance:** To monitor and record the rates of hospital infections.
- **Policy Making:** To develop protocols for sterilization, disinfection and waste management.

Functions of ICC

- **Training:** Organizing educational programs for staff on hand hygiene and PPE use.

- **Outbreak Investigation:** Identifying the source and controlling the spread if an infection outbreak occurs.
- **Environmental Monitoring:** Checking the cleanliness of OTs, ICUs and water supplies.
- **Antibiotic Policy:** Developing a "Restricted Antibiotic List" to ensure medicines are used only when necessary.

➤ **Role of Pharmacist in ICC**

The pharmacist acts as a bridge between the laboratory (Microbiology) and the bedside (Clinical care):

- **Antiseptic Management:** Ensuring the procurement and supply of high-quality disinfectants and hand rubs.
- **Sterilization Oversight:** Monitoring the chemicals used in the Central Sterile Supply Department (CSSD).
- **Data Analysis:** Providing reports on antibiotic consumption patterns to the committee.
- **Waste Management:** Ensuring the safe disposal of expired or cytotoxic drugs to prevent environmental contamination.

➤ **Role of Pharmacist in Preventing Antimicrobial Resistance (AMR)**

Antimicrobial Resistance occurs when bacteria/viruses no longer respond to medicines. Pharmacists play a critical "Gatekeeper" role:

- **Antimicrobial Stewardship (AMS):** Ensuring the right antibiotic is given for the right duration. They suggest de-escalation (switching from broad-spectrum to narrow-spectrum drugs) once lab results are available.
- **Prescription Audit:** Reviewing prescriptions to ensure they follow the hospital's antibiotic policy and checking for correct dosing.
- **Patient Counseling:** Educating patients on the importance of **completing the full course** of antibiotics, even if they feel better, to prevent the survival of resistant bacteria.
- **Prevention of Self-Medication:** Advising patients against using antibiotics for viral infections like the common cold or flu.
- **Monitoring ADRs:** Tracking adverse reactions to newer antibiotics to ensure their safety profile is maintained.

